JOINT ARMY-NAVY SPECIFICATION

HEXACHLORBENZENE

Army Number
5011-97A

Navy Number
51H3

This specification was approved by the War Department and the Navy Department for use of procurement services of the Army and the Navy and supersedes the following specifications.

U.S. Army
50-11-97
12 June 1941

Navy Department

A. APPLICABLE SPECIFICATIONS

A-1. The following specifications of the issue in effect on date of invitation for bids form a part of this specification:

U.S. ARMY SPECIFICATIONS

50-0-1 - General Specifications for Ammmunition except Small Arms Ammunition.[1]
100-2 - Standard Specification for Marking Shipments by Contractors.

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material.[2]

FEDERAL SPECIFICATION

RR-S-366 - Sieves; Standard, Testing.

B. GRADE

B-1. This specification covers one grade of hexachlorbenzene.

C. MATERIAL AND WORKMANSHIP

C-1. See section E.

D. GENERAL REQUIREMENTS

D-1. See section E.

E. DETAIL REQUIREMENTS

E-1. *Chemical and physical requirements.* - Hexachlorbenzene shall conform to the following chemical and physical requirements:

Color	White or cream
Moisture, percent (maximum)	0.20
Melting point, degree C. (minimum)	225
Benzene insoluble, percent (maximum)	0.10

[1]Applicable only to Army purchases.
[2]Applicable only to Navy Purchases.

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E-2. *Granulation.* - Hexachlorbenzene shall conform to the following granulation requirements, using U.S. Standard sieves conforming to the requirements of Federal Specification RR-S-366:

	Percent
Through No. 50 sieve (minimum)	99.0
Through No. 100 sieve (minimum)	95.0

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F-1. *Size of lots.* - Maximum, 5,000 pounds.

F-2. *Sampling.* - A minimum of 10 percent of the containers in the lot shall be selected by the Government inspector in such a manner as to be representative of the lot. When lots comprise less than 100 containers, either 10 containers or all containers in the lot shall be selected. Sufficient of the material to form a primary sample of 1 pound shall be removed from each selected container by means of a thief or a scoop. This sample shall be mixed thoroughly. A 4-ounce portion of the primary sample shall be placed in a tightly-stoppered bottle and labeled so that the container from which it was taken can easily be identified. The remaining portions of the primary samples shall be mixed together thoroughly and quartered until a composite sample of 1 pound is obtained. The composite sample shall be placed in a tightly-stoppered container and labeled to show the name of the material, manufacturer, plant, contract or order number, number of pounds in the lot, and lot number. All acceptance tests shall be made on the composite sample. The primary samples shall be held for possible future examination should the composite sample fail to meet the requirements of this specification.

F-3. *Inspection.* -

F-3a. *Army.* - Inspection shall be made in accordance with the requirements of U.S. Army Specification 50-0-1.

F-3b. *Navy.* - Inspection shall be at the point of delivery unless otherwise specified in the contract or order.

F-4. *Tests.* - The laboratory tests shall be made in accordance with the following paragraphs. For Navy purchases, the tests shall be made at a Government laboratory unless otherwise specified in the contract or order:

F-4a. *Color.* - Make visual examination for color.

F-4b. *Moisture.* - Transfer approximately 3 gm. of the sample to a tared glass weighing dish and weigh accurately. Dry the material for 48 hours in a vacuum desiccator and weigh. Calculate the loss in weight as percentage of moisture in the sample.

F-4c. *Melting point.* -

F-4c (1). *Apparatus.* - Set up a melting point bath equipped with a mechanical stirrer and a source of heat that can be easily regulated. A beaker of one to two liters capacity about three-fourths full of clean paraffin oil is suitable. Suspend an accurately standardized total immersion centigrade thermometer in the bath so that the bulb is not less than 1-1/2 inches from the bottom of the bath. If the mercury column will not be completely immersed at the temperature of the observed melting point, suspend a second thermometer about 1/2 inch from the first thermometer with the bulb approximately at the height of the middle of the exposed mercury column of the first thermometer.

F-4c (2). *Procedure.* - Grind a small portion (not over 5 gm.) of the sample so that it will pass through a U.S. Standard No. 100-mesh sieve, and dry for 2 hours at 100°C. or over sulfuric acid in a desiccator for 48 hours. Use thin walled capillary tubes of uniform diameter, long enough to extend above the top of the bath. Fill the tube with ground material to a depth of approximately 5 mm. compact by tapping, and fasten the tube to the standardized thermometer so that the lower end of the tube is in contact with the bulb of the thermometer. Start the stirrer, heat the bath rapidly to approximately 220°C, then gradually so that the rise in temperature does not exceed 1 degree in 3 minutes. Record the temperature observed at the instant the first meniscus of melted hexachlorbenzene appears across the capillary tube.

F-4c (3). *Report.* - If the mercury column is completely immersed at the melting point temperature, report this temperature with calibration corrections as the melting point of the sample. If part of the column is exposed, add the following correction to the observed melting point:

$$n\,(T\text{-}t) \times 0.000159$$

Where:

n = number of degrees in the exposed column.
T = uncorrected melting point.
t = average temperature of exposed mercury column.

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F-4d. *Benzene insoluble.* - Transfer a weighed portion of approximately 10 gm. of the sample to a 500-ml. beaker and dissolve in hot benzene. Filter the solution through a tared filtering crucible, rinsing out the beaker with benzene and passing the rinsings through the crucible. Wash the residue in the crucible thoroughly with hot benzene. Dry the crucible at 100°C. for 1 hour, cool in a desiccator, and weigh. Calculate the increase in weight as percentage of benzene insoluble matter.

F-4e. *Granulation.* - Fit a U.S. Standard No. 50 sieve on a U.S. Standard No. 100 sieve and attach a receiving pan to the bottom sieve. Place a weighed portion of 100 gm. of the sample on the upper sieve, and, by means of a soft brush, work as much of the material as possible through the sieve. Remove the No. 50 sieve and in the same manner work as much of the material as possible through the No. 100 sieve. Weigh the material remaining on each sieve and calculate the percentage of the sample passing through each sieve.

F-5. *Resubmission.* - If the composite sample representative of the lot fails to pass the inspection tests, the manufacturer shall have the option of having analysis of each primary sample made at no further expense to the Government. The manufacturer may then remove or replace defective portions of the lot represented by the primary samples which fail to meet the requirements, and submit the lot for acceptance, provided that the markings on the container are such that complete removal or replacement of defective portions of the lot can be made to the satisfaction of the Government inspector.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G-1. *Packing.* - Unless otherwise specified, hexachlorbenzene shall be delivered in standard commercial containers so constructed as to insure acceptance by common or other carriers, for safe transportation, at the lowest rate to the point of delivery.

G-2. *Marking.* - In addition to any special marking required by the contract or order, shipments for the Army shall be marked in accordance with the requirements of the U.S. Army Specification 100-2; for the Navy, in accordance with the requirements of the latest issue of the Navy Shipment Marking Handbook.

H. NOTES

H-1. Requests, requisitions, schedules and contracts or orders should contain the title of the specification, the number , and the date.

H-2. *Use.* - Hexachlorbenzene covered by this specification is intended for use in the manufacture of pyrotechnic compositions.

H-3. Copies of Joint Army-Navy specifications and Federal specifications (required for Army purchases) and U.S. Army specifications may be obtained as indicated in the "Index of United States Army. Joint Army-Navy, and Federal Specifications Used by the War Department". Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Agencies within the War Department will obtain copies of Joint Army-Navy, United States Army, and Federal specifications through established War Department channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H-4. Copies of Joint Army-Navy specifications and Federal specifications (required for Navy purchases), Navy Department specifications and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D.C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N.J. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H-5. Copies of this Joint Army-Navy Specification (required for Army purchases) may be obtained from the Office, Chief of Ordnance, Army Service Forces, Washington 25, D.C.

H-6. Copies of this Joint Army-Navy specification (required for Navy Purchases) may be obtained as indicated in paragraph H-4.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may be in any way related thereto.

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